

Multimodal Fake News Classification

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1 Introduction

The proliferation of misleading content on social media increasingly blends textual and visual cues, making single-modality detectors brittle in the presence of image–text mismatches, stylistic manipulations, and out-of-context media. Prior analysis in this project established that (i) text-only models provide strong baselines, (ii) lightweight vision models underperform on fine-grained categories, and (iii) naïve multimodal fusion can inherit weaknesses from either stream when encoders or training signals are insufficiently strong. These findings motivate a method that explicitly reasons about cross-modal consistency and addresses class imbalance while preserving the leak-safe principles adopted earlier.

2 Problem Statement and Design Overview

2.1 Task Definition

We tackle the task of *multimodal fake-news classification* on social-media posts combining short textual titles and associated images. Each post is labeled with one of $C \in \{2, 3, 6\}$ credibility classes. Formally, each sample $(x^{(t)}, x^{(v)}, y)$ consists of a tokenized title $x^{(t)}$, an RGB image $x^{(v)}$, and a target label y . The challenge lies in jointly modeling (i) semantic cues in concise, informal text, (ii) visual content that may complement or contradict it, and (iii) cross-modal alignment under imbalance or missing inputs. We therefore learn a function $f : (x^{(t)}, x^{(v)}) \rightarrow \hat{y}$ that maximizes macro-F1 while remaining robust when either modality is weak or absent. To achieve this, we adopt a *gated multimodal network* with *modality dropout* (*ModDrop*) that dynamically adjusts modality importance during training.

2.2 Model Goals

The model jointly leverages text and vision to produce context-aware predictions, handles text–image inconsistencies through adaptive gating and ModDrop, and improves minority-class performance in 3- and 6-way setups via balanced sampling and macro-F1-oriented evaluation.

2.3 Design Principles and Constraints

Our design follows three principles: encoder efficiency, imbalance awareness, and continuity and fairness. Pretrained backbones—ResNet-18 for images and a bidirectional GRU for text—provide expressive yet efficient representations. The pipeline maintains balanced split caps and uniform sampling to mitigate label and modality skew, emphasizing macro-level metrics over accuracy. Finally, identical class splits and preprocessing steps are retained from prior parts to ensure fair, leak-safe comparison focused solely on the impact of the proposed gating and ModDrop mechanisms.

2.4 Workflow Overview

To bridge preprocessing and modeling, [Figure 1](#) summarizes the end-to-end pipeline implemented in `1.merging_data.py`, `2.text_preprocessing.py`, and `3.train_gated.moddrop.py`.

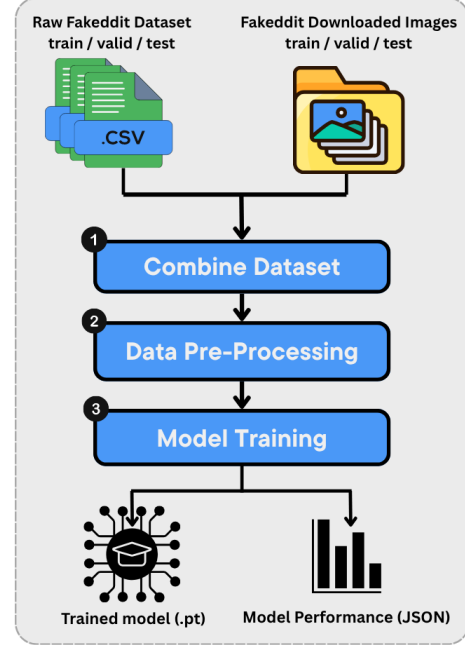


Figure 1: Overall workflow showing dataset combination, preprocessing, and model training pipeline.

3 Dataset and Preprocessing

3.1 Dataset and Splits

We employ the Fakeddit multimodal benchmark, which provides post-level metadata, short textual titles, and corresponding images. The raw train, validation, and test TSVs are first cleaned and merged into consistent CSV manifests using `1.merging_data.py`.

The merging step performs the following operations:

- Attaches a string `hasImage` flag and relative `image_path`;
- Verifies that images exist locally and discards missing or corrupted entries;
- Prioritizes paired (text + image) samples, then fills up to the split cap with text-only entries;
- Caps each split to fixed sizes (50k/5k/5k) for train, validation, and test to ensure comparability across experiments.

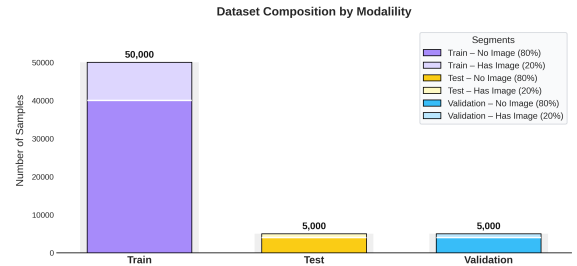


Figure 2: Dataset Composition by Modality. Roughly 20% of entries include both modalities. This imbalance motivates the gated ModDrop mechanism to adaptively handle missing or weak modalities.

The final manifests (`train_final.csv`, `valid_final.csv`,

test_final.csv) are accompanied by a stats.json file summarizing total and modality-specific counts. This setup preserves the overall class distribution and modality proportions established in earlier project stages.

Overall, this data preparation phase guarantees that each modality is consistently represented, and that label proportions and image–text relationships remain comparable with earlier baselines.

3.2 Text Preprocessing

Text normalization is handled by 2_text_preprocessing.py, which prepares the titles for the textual encoder. For each record, the fields clean_title and title are concatenated, providing both a raw and a cleaned view of the post text. The cleaning function lowercases text, removes URLs, filters non-alphabetic characters, and standardizes whitespace. Rows missing both fields are discarded.

Tokens are then extracted via whitespace splitting and optionally processed with NLTK stopwords removal and lemmatization (executed only if NLTK data are available). A Keras Tokenizer with a 120,000-word vocabulary limit is fit on all splits (train, valid, test) without an explicit oov_token, meaning unseen words map to index 0.

Each title is encoded as an integer sequence and post-padded or truncated to a uniform length of $L = 15$. The script saves:

- tokenized arrays (train_seq.npy, valid_seq.npy, test_seq.npy);
- aligned CSVs (train_used.csv, valid_used.csv, test_used.csv);
- tokenizer.json for reproducibility; and
- a meta.json summarizing vocabulary size, sequence length, label column, and sample counts.

This produces compact and consistent input sequences suitable for the GRU-based encoder in the multimodal model, ensuring reproducibility and fair comparison with prior text-only baselines.

3.3 Image Preprocessing

Image preparation is handled dynamically during training within 3_train_gated_moddop.py. Each image is loaded using PIL, converted to RGB, and standardized through resizing and center-cropping to a uniform spatial resolution of 224×224 pixels, ensuring consistent input dimensions across batches. Normalization follows the statistical distribution of the ImageNet dataset, aligning input intensities with those expected by the pretrained ResNet-18 backbone. This alignment facilitates stable gradient flow and faster convergence by keeping feature activations within the range for which the encoder’s weights were originally optimized.

When an image is missing or unreadable, it is replaced with a zero-valued tensor and the corresponding hasImage flag is set to 0. This design guarantees shape consistency and allows the gated fusion network to handle absent modalities gracefully. Data augmentation is intentionally minimal and deterministic—limited to geometric and intensity standardization—to preserve comparability across experimental runs and avoid introducing modality-specific bias.

3.4 Final Output

After preprocessing, the project directory contains verified CSV manifests, tokenized text arrays, and standardized image tensors ready for the multimodal dataloaders.

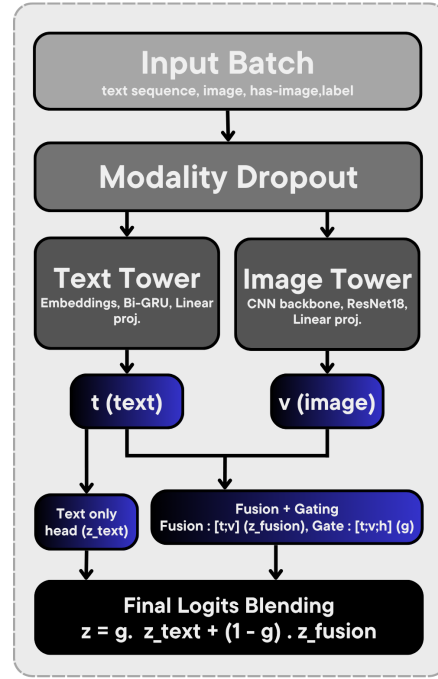


Figure 3: Overview of the proposed gated late-fusion network enhanced with modality dropout.

4 Model Architecture

As illustrated in Figure 3, our architecture adopts a gated late-fusion network enhanced with modality dropout (ModDrop) to dynamically regulate the contribution of text and image information. The model integrates three main components: a bidirectional GRU-based text encoder, a pretrained ResNet-18 visual encoder, and a fusion module governed by a learnable gating mechanism. This design extends the earlier GRU–CNN baseline by introducing (i) a stronger pretrained visual backbone for robust feature extraction, (ii) explicit modality-level dropout for regularization, and (iii) an adaptive gating network that interpolates between unimodal and multimodal representations depending on input availability and reliability.

4.1 Text Tower

The text tower transforms tokenized sequences into dense semantic embeddings that capture linguistic nuances and contextual relationships. Each token index is mapped to a 200-dimensional trainable embedding vector and passed through a bidirectional GRU with hidden size 256. This setup enables the model to capture both forward and backward dependencies in short, often noisy, social-media text. The concatenation of the final forward and backward states forms a 512-dimensional text representation $\mathbf{t} \in \mathbb{R}^{512}$. A dropout layer is applied after the GRU to mitigate overfitting. This representation serves a dual purpose: it is fed to a *text-only head* for unimodal classification, and simultaneously forwarded to the fusion and gating module for multimodal reasoning. By preserving a distinct text-only prediction branch, the architecture retains interpretability—allowing analysis of how much the final decision depends purely on textual cues.

4.2 Image Tower

The image tower extracts visual features using a ResNet-18 backbone pretrained on ImageNet. The network is truncated before its classification layer, preserving convolutional filters that encode color,

texture, and object-level patterns. These pre-trained filters provide strong inductive biases that transfer effectively to multimodal fake-news data. The global average-pooled output is projected through a linear layer into a 512-dimensional embedding $\mathbf{v} \in \mathbb{R}^{512}$, aligned dimensionally with the text representation. To stabilize training and limit parameter growth, the convolutional backbone remains frozen, while only the projection layer is trained. This allows efficient adaptation to the dataset’s image domain while avoiding overfitting to small sample sizes. Normalization and resizing operations from preprocessing ensure consistency with ImageNet statistics, improving convergence and stability.

4.3 Fusion and Gating Mechanism

Once the modality-specific embeddings are obtained, they are integrated through a combination of fusion and gating. First, the text and image embeddings are concatenated $[\mathbf{t}; \mathbf{v}]$ and processed through a two-layer fully connected fusion MLP ($1024 \rightarrow 256 \rightarrow \#classes$) with ReLU activation and dropout, producing fused logits $\mathbf{z}_{fusion}$. Meanwhile, the text-only head produces its own unimodal logits $\mathbf{z}_{text}$.

A lightweight gating network receives the concatenated vector $[\mathbf{t}; \mathbf{v}; h]$, where h is the binary `hasImage` flag, and outputs a scalar gate $g \in [0, 1]$ through a sigmoid activation. The gate modulates the final prediction via:

$$\mathbf{z} = g \mathbf{z}_{text} + (1 - g) \mathbf{z}_{fusion}. \quad (1)$$

Intuitively, when g is close to 1, the model primarily trusts textual features—useful when images are irrelevant or misleading. Conversely, when g is smaller, the network incorporates visual evidence more strongly. This adaptive blending mechanism enables a smooth, data-driven interpolation between unimodal and multimodal reasoning, supporting robust decision-making under varying modality reliability.

4.4 Modality Dropout (ModDrop)

To prevent overfitting and encourage redundancy across modalities, entire modality streams are randomly masked during training. Text sequences are replaced with zero-padding with probability $p_{text} = 0.1$, and image tensors are replaced with zero-valued arrays with probability $p_{image} = 0.3$. The `hasImage` flag ensures that naturally missing images are always treated as dropped. This dropout is applied at the batch level and disabled at inference time. By occasionally silencing an entire modality, the model learns complementary feature representations in each tower and becomes more robust to real-world scenarios where one channel may be unreliable or unavailable.

4.5 Architectural Rationale

This architecture is designed to balance *expressivity*, *robustness*, and *efficiency*. The pretrained ResNet-18 backbone stabilizes visual learning and provides transferable low-level features. The bidirectional GRU captures contextual dependencies and semantic nuance in short, noisy social-media titles. The fusion and gating module introduces adaptability, allowing the model to modulate attention between modalities according to their reliability signals. Finally, ModDrop regularization ensures modality independence and prevents feature co-adaptation. Together, these components yield a cohesive and interpretable multimodal model that performs effectively under both complete and partial modality conditions.

Together, these elements form a balanced architecture that is both interpretable and resilient. It maintains competitive performance

under complete multimodal conditions while gracefully handling missing or corrupted inputs. This makes the Gated ModDrop model particularly suitable for real-world misinformation detection tasks, where multimodal completeness and quality can vary drastically.

5 Experimental Setup and Reproducibility

To ensure full reproducibility, the complete dataset (including images, metadata, and preprocessing scripts) is publicly available at our Github repository ([link](#)). The dataset follows the structure of the Fakeddit dataset, containing paired text and image samples divided into training, validation, and testing subsets.

To reproduce the experiments, create a Python 3.10 virtual environment and install the required dependencies. Major libraries include PyTorch, Torchvision, TensorFlow, Keras, Transformers, Scikit-Learn, Pandas, NumPy, NLTK, and Pillow. A complete list of packages with their versions is provided in the accompanying `requirements.txt` file.

After setting up the environment, update the `DATA_ROOT` variable in each of the three scripts—dataset combination, preprocessing, and model training—to match the path of your dataset folder. This ensures that all scripts correctly locate and use the data for training and evaluation.

Finally, modify the `TEXT_SUBDIR`, `NUM_CLASSES`, and output file names (`.pt` and `.json`) in the model training script depending on how many classes (2-way, 3-way, or 6-way) the model should be trained for.

6 Results

Table 1 summarizes the performance across all granularities. The model demonstrates consistent and robust behavior, achieving strong accuracy and macro-F1 scores even as the task complexity increases.

Table 1: Performance of the Gated ModDrop model across different label granularities.

Task	Accuracy	Loss	Macro-F1
2-Way	0.8222	0.7285	0.8214
3-Way	0.8064	0.7717	0.7266
6-Way	0.7052	1.2216	0.5182

The results indicate that while accuracy and F1-score naturally decline with finer-grained labeling, the model remains capable of learning meaningful discriminative features. This consistency highlights the role of the *gating mechanism* in balancing contributions from text and image modalities, while *ModDrop regularization* improves robustness against missing or noisy inputs.

7 Conclusion

We introduced a Gated ModDrop Network for multimodal fake-news detection that integrates adaptive fusion with modality-level dropout. By dynamically adjusting the reliance on text and image modalities, the model achieves balanced and interpretable predictions across all three classification granularities. Even in the challenging six-class setting, it maintains stable precision–recall trade-offs, confirming the effectiveness of the architecture.

Overall, these findings emphasize that adaptive gating combined with modality dropout fosters more resilient multimodal misinformation classifiers, capable of maintaining strong generalization across diverse fake-news detection scenarios.